

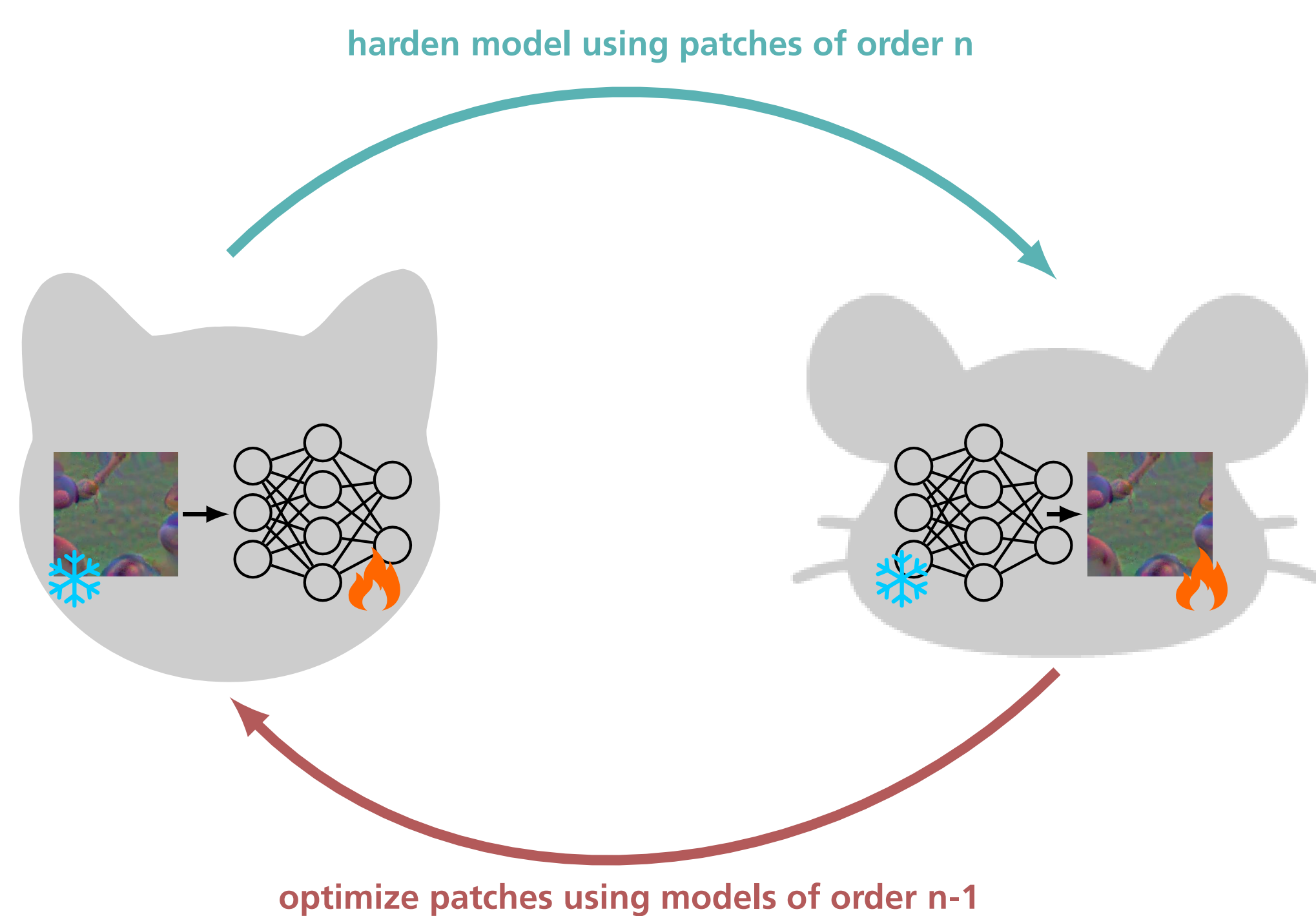
Higher-Order Adversarial Patches for Real-Time Object Detectors

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Motivation

The Problem

Adversarial patches can fool object detectors in the physical world. Adversarial training hardens a model against them - but only until an attacker optimizes a new patch against that hardened model. This cat-and-mouse dynamic can be replayed indefinitely.



Key Question

To what extent does repeated play of this cat-and-mouse game strengthen the models - and the patches?

Definition — "Order"

The **order of a patch/model** = **number of cat-and-mouse cycles played**. A 1st order patch fools a standard (0th order) detector; an (n+1)th order patch is crafted to bypass an nth order hardened detector.

Method & Setup

Setup

- Attack type: vanishing attack.
- Target class: Person.
- Detector: YOLOv10b [1].

Adversarial Patches

- Trained on *INRIA Person* [2].
- Resolution of 256 × 256 pixels.
- Optimized over 150 epochs using *AdamW* [3].
- Initial Learning rate of 0.01 reduced every 50th epoch.
- Augmented using color jitter, random rescaling, rotation and perspective transformation.
- Target: Pre-sigmoid prediction confidences.

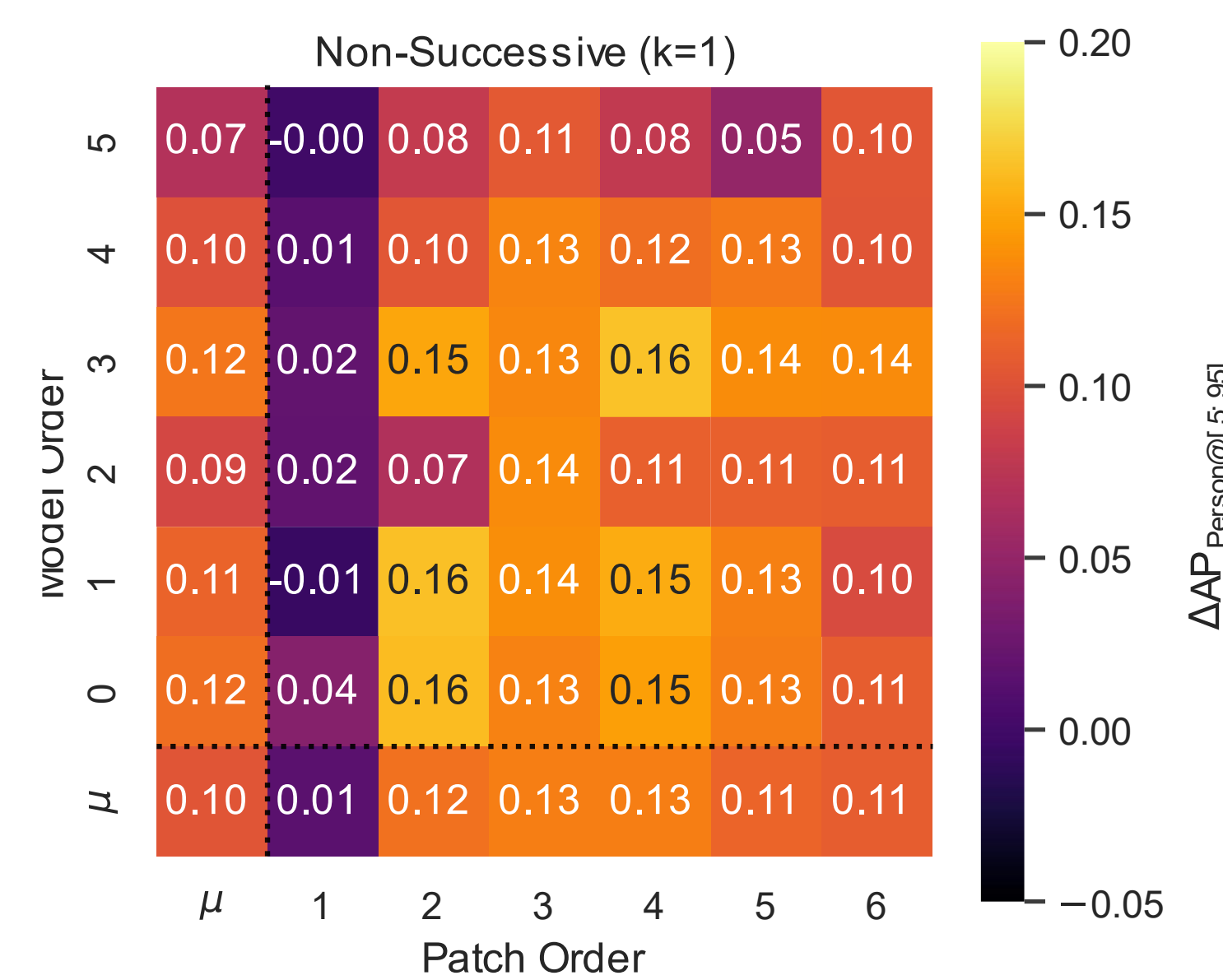
Adversarial Training

- Trained on *COCO* [4].
- Random adversarial patches are placed within ground-truth bounding boxes of Persons with a probability of 25%.
- The only augmentation of these patches is a resize.

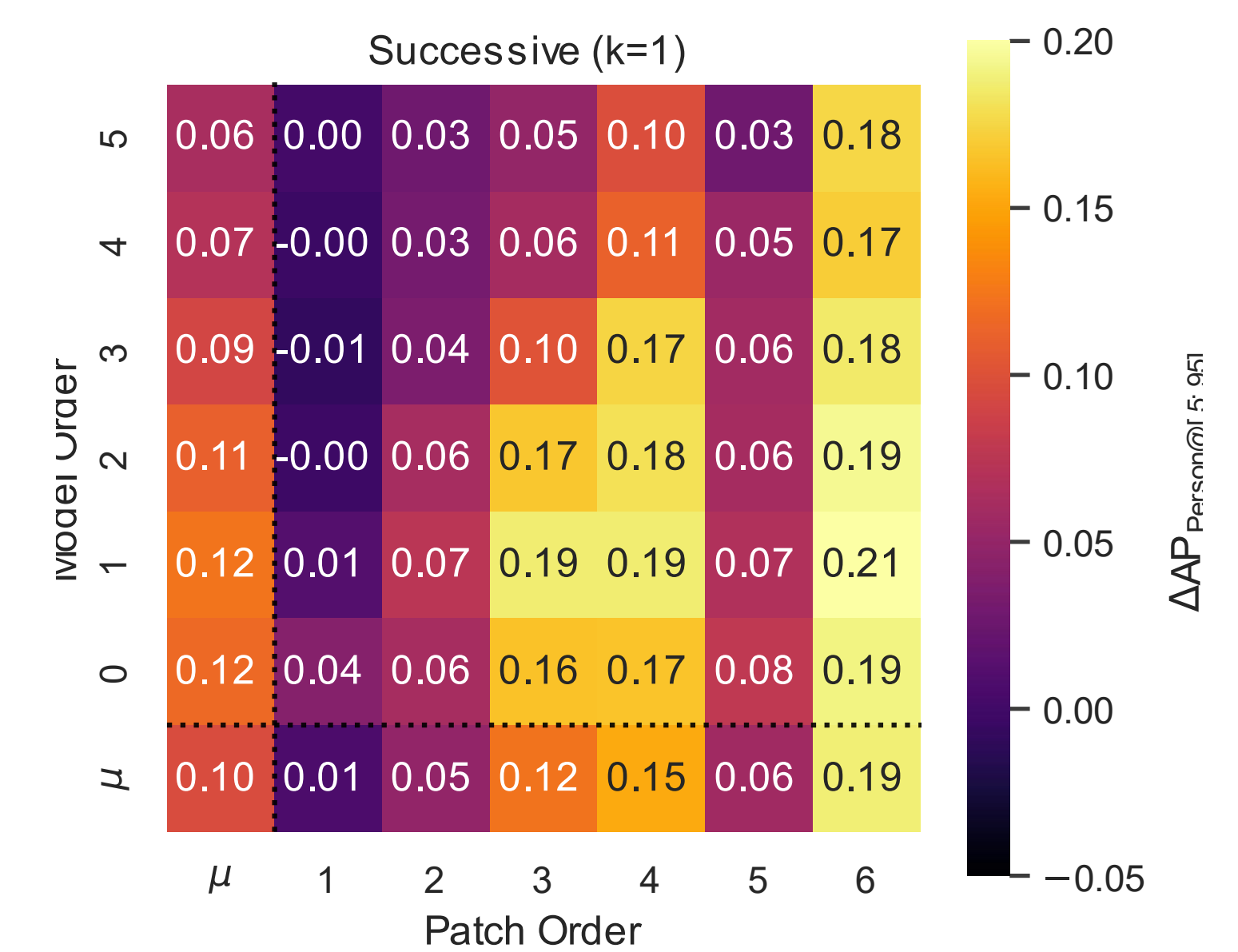
Results

Higher-Order Attacker-Defender Dynamics

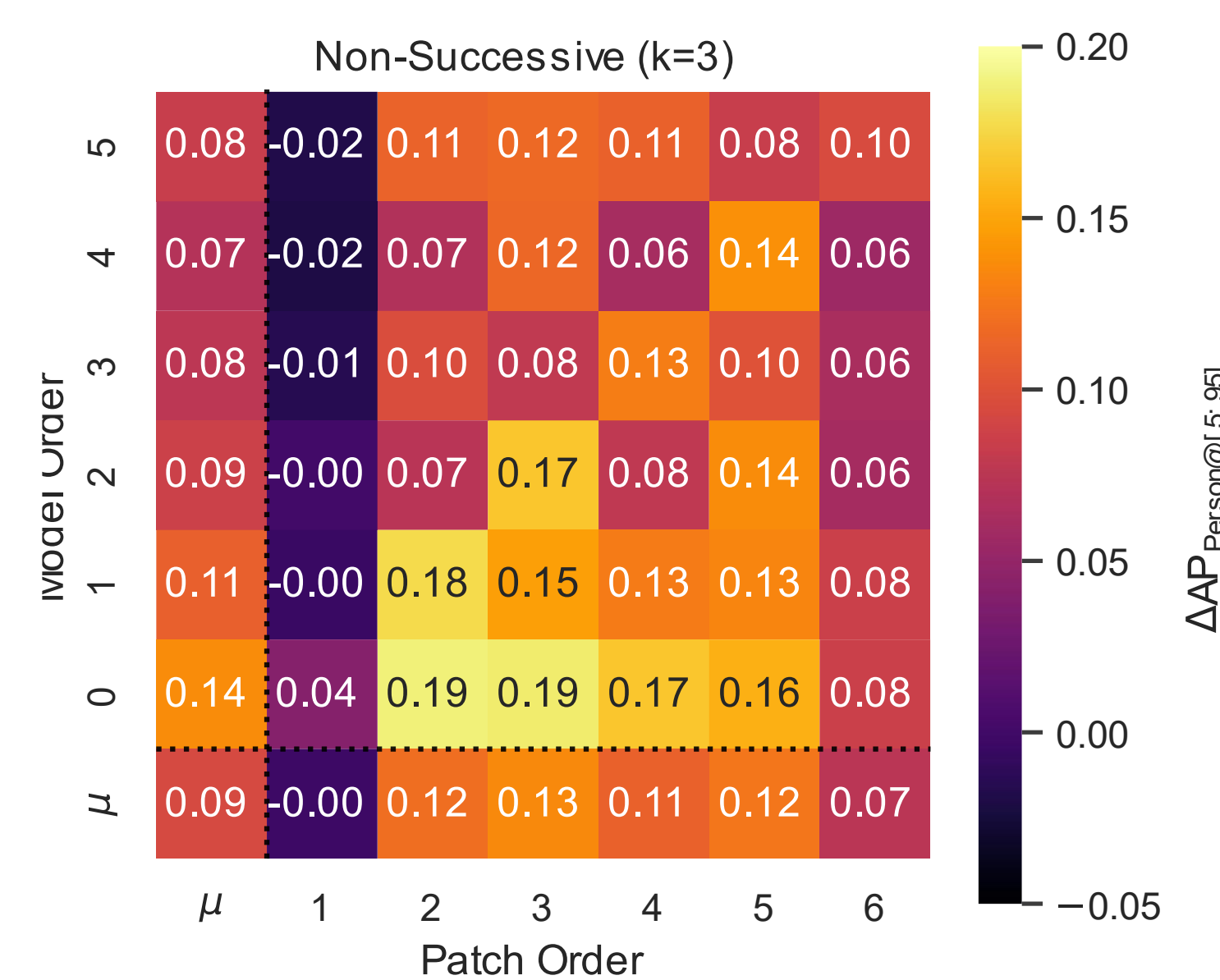
Attack impact. Each cell = mean ΔAP for a network order (row) attacked by a patch order (column). The darker the shade, the stronger the attack.



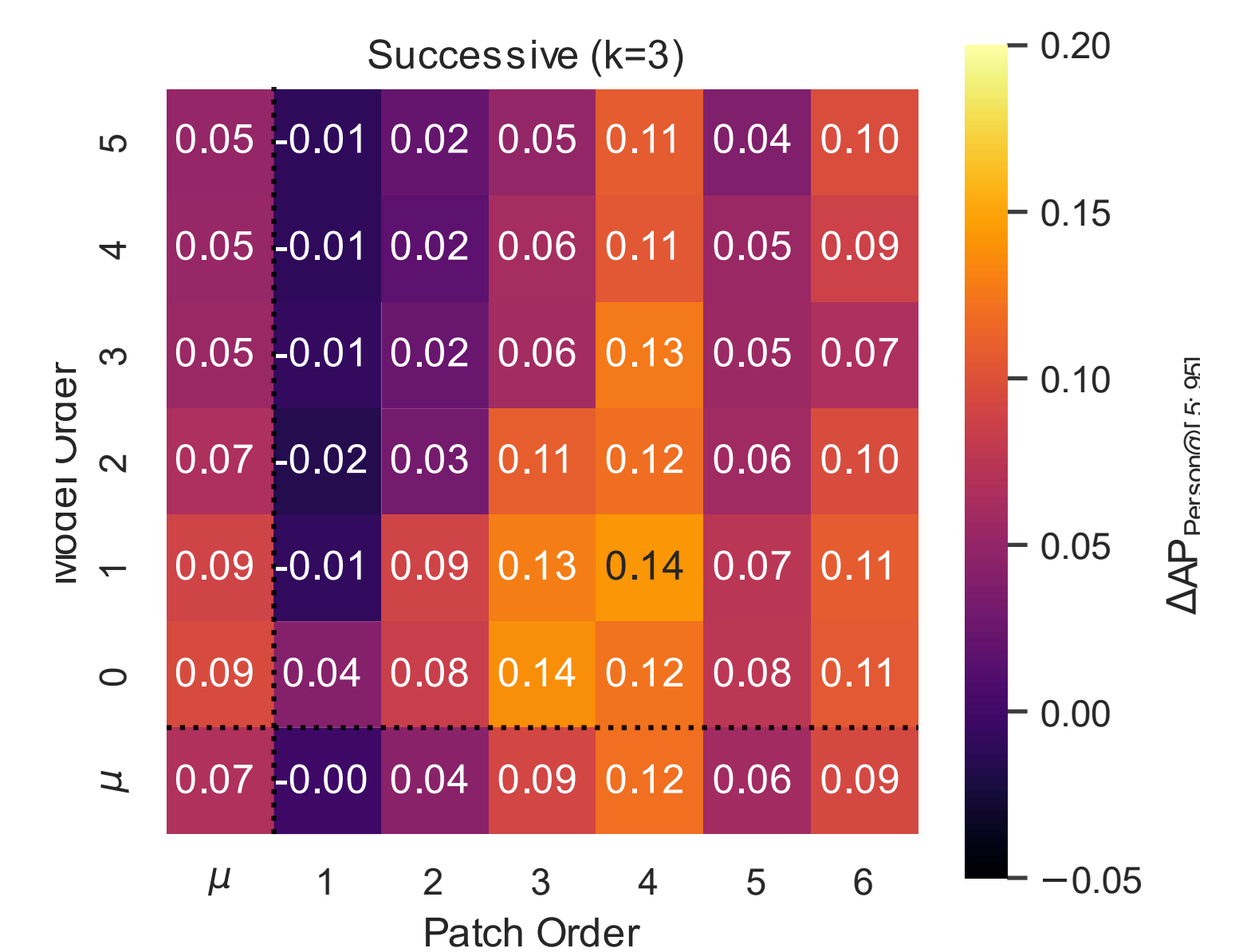
k=1 patch optimized, **non-successive** training: The nth order model is exposed to the latest single nth order patch. A single new patch (k=1) is optimized each order.



k=1 patch optimized, **successive** training: The nth order model is exposed to all ($\leq n$)th order patches. A single new patch (k=1) is optimized each order.

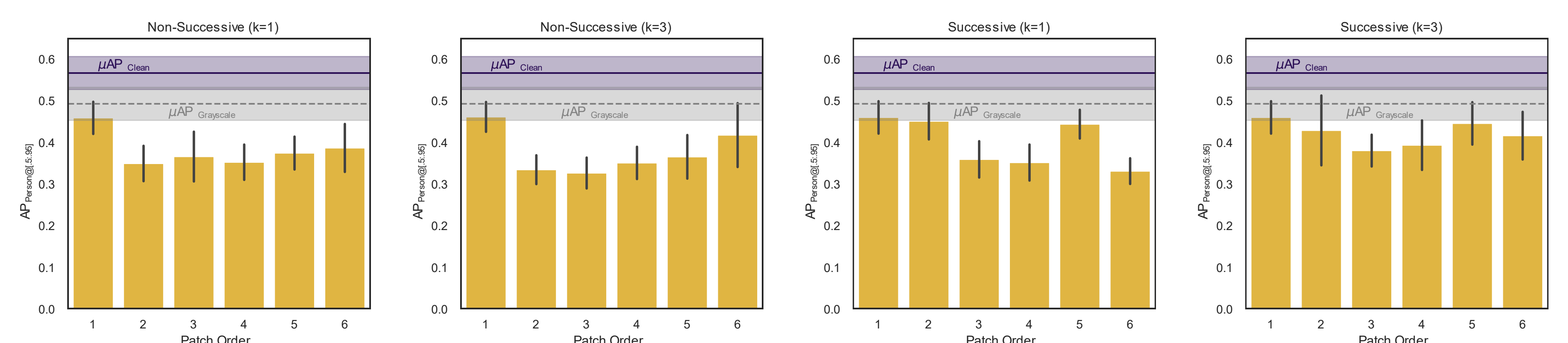


k=3 patches optimized, **non-successive** training: The nth order model is exposed to all nth order patches. Multiple new patches (k=3) are optimized each order.



k=3 patches optimized, **successive** training: The nth order model is exposed to all ($\leq n$)th order patches. Multiple new patches (k=3) are optimized each order.

Transferability



Patches transfer across 21 unhardened YOLOv9/10/11/12 [1] detectors. Successive training yields weaker but more general patches; non-successive yields stronger, less transferable ones.

Key Findings

- Higher-order patches hit harder and generalize better than 1st order patches - even against detectors they were never trained on.
- Higher-order detectors are more robust, but performance is only partly restored - never back to clean level.
- Adversarial training alone is not enough to harden a detector against this attack class (attack-generalization gap [5]).

Future Work

Does the cat-and-mouse game converge or diverge? Accumulation points could guide fundamentally more robust hardening.

References

- [1] Jocher et al.: "*Ultralytics YOLO*", 2024.
- [2] Dalal & Triggs: "*Histograms of oriented gradients for human detection*", CVPR, 2005
- [3] Loshchilov & Hutter: "*Decoupled Weight Decay Regularization*", ICLR, 2019
- [4] Lin et al.: "*Microsoft coco: Common objects in context*", ECCV, 2014
- [5] Cai et al.: "*Curriculum Adversarial Training*", IJCAI, 2018

